

Omnichannel Customer Intelligence Platform

Fashion & Retail — ML Insights Report

Auto-generated · May 2026 · Senior ML Engineering Project
XGBoost · SHAP · KMeans · A/B Testing · MLflow

8,000	€3.63M	38.4%	37.2%
Customers Analysed	CLV Portfolio (12m)	High Churn Risk	VIP Customers

This report presents a comprehensive analysis of 8,000 fashion retail customers across an omnichannel ecosystem (e-commerce, mobile app, physical stores, and social commerce). Three machine learning models were trained and validated: a **Churn Prediction** classifier, a **Customer Lifetime Value** regressor, and a **Customer Segmentation** model. Results are enriched with SHAP-based explainability and statistically validated A/B tests.

1. Executive Summary

The Fashion Intelligence Pipeline reveals a customer base under significant churn pressure, with **38.4% of customers at high risk** of abandonment — yet simultaneously showing strong value concentration: the top 37.2% of customers (VIP segment, CLV > €400) account for the majority of the €3.63M 12-month revenue forecast. This tension between retention risk and value concentration defines the central strategic challenge.

Metric	Value	Status
Total customers analysed	8,000	■
High churn risk (P > 60%)	3,073 (38.4%)	■ Action required
VIP customers (CLV > €400)	2,975 (37.2%)	■ Protect
Total CLV portfolio (12 months)	€3,627,030	■
Customers needing urgent action	1,938 (24.2%)	■ Urgent

■ **Insight:** 1,938 customers simultaneously have high churn risk AND high CLV. Retaining even 30% of these at an average CLV of €450 would protect ~€261K in annual revenue — making personalised retention campaigns the highest-ROI investment available.

CRM Action Distribution

The Action Matrix (Churn × CLV) automatically segments the base into four operational priorities:

Action	Customers	% of Base	Avg CLV Est.	Priority
■ Urgent Retention	1,938	24.2%	> €200	P1 — This week
■ Nurture VIP	1,806	22.6%	> €400	P2 — Monthly
■ Reactivation	1,884	23.5%	Mixed	P3 — Automated
■ Maintenance	2,372	29.6%	< €200	P4 — BAU

■ **Insight:** The Urgent Retention (P1) and Nurture VIP (P2) segments together represent 46.8% of the customer base but likely 70%+ of total revenue. These 3,744 customers should be the focus of all high-touch CRM efforts.

2. Churn Prediction Model

The XGBoost Classifier was trained on 25 features derived from demographic, behavioural, and transactional data. Class imbalance was addressed via **scale_pos_weight** calibration. The model outputs a continuous churn probability score for each customer, enabling fine-grained CRM prioritisation.

Model Performance

Metric	Value	Benchmark	Interpretation
ROC-AUC	0.7956	> 0.75 = Good	For any (churner, stayer) pair, the model ranks correctly 79.6% of the time
PR-AUC	0.6108	> 0.35 = Good	Strong precision-recall balance despite class imbalance
Threshold	0.60	Business-calibrated	Balances false alarm cost vs missed churn cost

Business Interpretation of ROC-AUC = 0.7956

A ROC-AUC of 0.80 means that if you randomly select one customer who churned and one who stayed, the model will correctly identify the churner as higher-risk **in 4 out of 5 cases**. This is strong enough for production CRM use — industry benchmarks for retail churn models typically range from 0.70 to 0.85.

Score Distribution Insight

The model produces a bimodal score distribution — customers who stay cluster near 0.10–0.25, while churning customers cluster near 0.65–0.90. This separation is the key signal: customers with scores above 0.60 are **3,073 individuals** where intervention is statistically justified.

■ **Insight:** At threshold 0.60, the model identifies the 38.4% most at-risk customers. A targeted retention campaign reaching even 20% of these with a 15% success rate would save approximately 92 customers — worth ~€41,400 in annual CLV.

3. Churn Drivers — Explainable AI (SHAP)

SHAP (SHapley Additive exPlanations) decomposes each prediction into the exact contribution of every feature, satisfying the **additivity property**: the sum of all SHAP values equals the difference between the prediction and the population average. This makes SHAP values directly actionable — each feature maps to a business lever.

Top 5 Churn Drivers

Rank	Feature	mean SHAP	Direction	Business Action
1	Days since last purchase	0.7241	↑ days = ↑ churn	Trigger win-back campaign at 60/90/180-day marks
2	Support tickets (12m)	0.3017	↑ tickets = ↑ churn	Proactive service recovery for customers with 2+ tickets
3	Recency score	0.2696	↓ score = ↑ churn	Recency score < 0.01 → automated re-engagement flow
4	Risk index	0.2622	↑ index = ↑ churn	Composite score — monitor weekly for early warning
5	Email open rate	0.1200	↓ rate = ↑ churn	Customers with < 5% open rate need channel switch to push/SMS

Deep Dive: Days Since Last Purchase (SHAP = 0.7241)

This single feature accounts for the **largest share of churn prediction power**, with a mean absolute SHAP value more than twice the next feature. The dependence plot reveals a non-linear relationship: churn risk accelerates sharply after 90 days and plateaus near 1.0 after 270 days. This implies three distinct intervention windows:

Window	Days Inactive	Churn Probability	Recommended Action
Early warning	60–90 days	~35–45%	Personalised product recommendation email
High alert	91–180 days	~55–70%	Discount voucher + personal outreach
Critical	180+ days	~75–90%	Win-back campaign + loyalty tier review

Support Tickets as a Leading Indicator

With SHAP = 0.3017, unresolved service friction is the **second strongest churn predictor**. Customers with 3+ support tickets in 12 months are disproportionately represented in the high-risk segment. This points to a service recovery gap: these customers are giving the brand repeated chances to fix the relationship before leaving. A proactive outreach programme (personal call or guaranteed resolution SLA) at the 2-ticket mark could convert service dissatisfaction into loyalty.

■ **Insight:** The top 2 SHAP features (recency + support) are both lagging indicators of disengagement. Monitoring email open rate (SHAP = 0.12) as an early-warning signal — before recency deteriorates — allows intervention at lower cost and higher success rate.

4. Customer Lifetime Value Forecast

The XGBoost Regressor forecasts each customer's 12-month gross contribution margin, defined as: **CLV = avg_order_value × total_purchases × 0.15 + noise**. The 15% margin assumption is a conservative estimate for mid-market fashion retail; actual margins should be substituted for production deployment.

Model Performance

Metric	Value	Interpretation
R²	0.9898	98.98% of CLV variance explained — near-perfect fit
MAE	€40.27	Average prediction error of €40 per customer
RMSE	€52.54	Penalises large errors — still under €53 on average

Portfolio Analysis

Total predicted CLV for the 8,000-customer base is **€3,627,030 over 12 months**. This figure represents the gross contribution margin expected if churn rates remain unchanged. Given that 38.4% of customers are at high churn risk, the **at-risk portfolio** — the CLV that could be lost to churn — is estimated at approximately €1.4M.

Segment	Customers	Est. CLV (12m)	Revenue at Risk	Action
High churn risk (P > 60%)	3,073	~€1.4M	€1.4M at risk	■ Retention campaign
VIP (CLV > €400)	2,975	~€1.8M	Protect	■ Loyalty programme
Stable base	1,952	~€400K	Low risk	■ Standard CRM

■ **Insight:** A retention programme with a 20% success rate on the 1,938 Urgent Retention customers (avg CLV ~€350) would recover approximately €135,660 in annual revenue — a meaningful return on a modest CRM investment.

5. Customer Segmentation

KMeans clustering on 9 behavioural features (RFM + engagement + omnichannel) identified **K=2 as the optimal number of clusters** via Silhouette Score maximisation. While K=2 may seem parsimonious, the Silhouette Score of 0.2210 confirms that two genuinely distinct behavioural archetypes dominate the customer base.

Silhouette Score Interpretation

Score Range	Interpretation	Our Score
> 0.50	Excellent — well-separated clusters	—
0.35–0.50	Good — clear separation	—
0.20–0.35	Reasonable — overlapping real-world data	0.2210 ■
< 0.20	Weak — consider different approach	—

A Silhouette Score of 0.22 is typical for customer behavioural data, where segments overlap by design — customers do not fall into perfectly distinct groups. The score validates that the two-cluster solution is meaningful and not arbitrary.

Segment Profiles

Dimension	Segment A — Active / High Value	Segment B — At Risk / Passive
Typical recency	< 60 days	> 120 days
Purchase frequency	High (10+ orders)	Low (1–3 orders)
Engagement index	> 0.40	< 0.20
Omnichannel usage	Multi-channel	Single channel
Churn risk	Low (< 25%)	High (> 50%)
CLV profile	VIP / Champions	At Risk / Reactivation
CRM strategy	Loyalty + upsell	Win-back + re-engage

■ **Insight:** The binary segmentation (Active vs At-Risk) is operationally powerful: it maps directly to two CRM journeys — a loyalty deepening track for Segment A and a re-engagement track for Segment B. For richer micro-segmentation, increasing K to 4–5 with additional features (brand affinity, category preference) is recommended in Phase 2.

6. A/B Testing — Campaign Validation

Three randomised controlled experiments were conducted using a two-sided Z-test of proportions with significance threshold $\alpha = 0.05$. Each test measured conversion rate differences between control and variant groups across channels and offers.

Campaign	Control	Variant	Lift	p-value	Sig ?	Winner
Email vs Push Notification	Email (ctrl)	Push Notif.	+30.4%	0.0027	■	Push Notification
Discount 10% vs Free Shipping	Discount 10%	Free Shipping	+14.6%	0.0801	■	Not significant
Instagram vs TikTok Ads	Instagram	TikTok Ads	+48.6%	0.0079	■	TikTok Ads

Test 1: Push Notification Wins (+30.4%, p=0.0027)

Push notifications significantly outperformed email for customer reactivation. At $p=0.0027$, this result is robust — there is only a 0.27% chance the lift is due to random variation. **Recommendation:** migrate reactivation campaigns from email-first to push-first for inactive customers (last purchase 60–120 days ago). Email remains valuable for VIP nurturing where open rates are high.

Test 2: Free Shipping vs Discount — Inconclusive (p=0.0801)

Despite a +14.6% observed lift for Free Shipping, the result did not reach statistical significance. This does NOT mean the offers are equivalent — it means we lack sufficient evidence at this sample size. **Recommendation:** run a follow-up test with $n=3,500$ per group (power analysis suggests this achieves 80% power at $\alpha=0.05$ for a 14% lift). Additionally, segment by basket size — free shipping may outperform for high-AOV customers while discounts work better for price-sensitive segments.

Test 3: TikTok Ads Win (+48.6%, p=0.0079)

TikTok Ads significantly outperformed Instagram for Gen Z acquisition, with a substantial +48.6% lift in conversion. At $p=0.0079$, this is a statistically robust result. **Caution:** both absolute conversion rates remain low (approximately 3–4%), suggesting social commerce acquisition is still early-stage. **Recommendation:** reallocate 30–40% of Instagram social budget to TikTok, but monitor LTV of acquired cohort before full reallocation — low-cost acquisition does not always translate to high-quality customers.

■ **Insight:** 2 out of 3 tests returned significant results. The combined findings suggest a channel mix shift: push notifications for reactivation, TikTok for acquisition. The inconclusive offer test (Discount vs Free Shipping) should be re-run with a larger sample before budget decisions are made.

7. Strategic Recommendations

The following recommendations synthesise findings from all three models and the A/B tests into a prioritised action plan, ordered by estimated revenue impact.

Priority 1 — Urgent Retention Programme (1,938 customers)

- **Target:** All customers with $P(\text{churn}) > 0.60$ AND $CLV > €200$
- **Channel:** Push notification (validated by A/B Test 1, +30.4% lift)
- **Trigger:** Automated at 90-day inactivity mark (SHAP insight: highest inflection point)
- **Offer:** Personalised win-back voucher (test Free Shipping vs Discount in follow-up A/B)
- **Revenue impact:** 20% success rate → ~€135K recovered per cohort
- **Timeline:** Launch within 2 weeks

Priority 2 — VIP Nurturing Programme (1,806 customers)

- **Target:** $CLV > €400$, churn risk $< 30\%$ — the most valuable stable segment
- **Goal:** Increase purchase frequency and deepen omnichannel engagement
- **Actions:** Exclusive early access, loyalty tier upgrades, personalised styling content
- **Channel:** Email (high open rate in this segment) + in-app personalisation
- **Revenue impact:** 10% frequency uplift on €1.8M segment = +€180K annually

Priority 3 — Service Recovery (high-ticket customers)

- **Target:** Any customer with 2+ support tickets in 12 months (SHAP #2 churn driver)
- **Intervention:** Proactive personal outreach before the 3rd ticket
- **Goal:** Convert service friction into loyalty — customers who get resolved issues often become advocates
- **Metric:** Track reduction in churn rate within this cohort vs control

Priority 4 — Channel Mix Rebalancing

- **Reactivation:** Migrate to push-first (A/B Test 1 — +30.4% vs email)
- **Acquisition:** Increase TikTok budget (A/B Test 3 — +48.6% vs Instagram)
- **Offer testing:** Re-run Discount vs Free Shipping with $n=3,500+$ per group
- **Email:** Retain for VIP nurturing and loyalty communications where open rates are high

Consolidated Impact Estimate

Initiative	Investment Level	Est. Revenue Impact	Confidence
Urgent Retention Programme	Medium	+€135K/cohort	High (model + A/B validated)
VIP Nurturing Programme	Low	+€180K/year	Medium (model validated)
Service Recovery	Low	+€50–80K/year	Medium (SHAP validated)
Channel Mix Rebalancing	Low	Cost reduction	High (2/3 A/B significant)

8. Tech Stack & MLOps

Model Registry — MLflow

Experiment	Run ID (prefix)	Key Metric	Status
fashion-churn-prediction	027bcb09aa87...	AUC = 0.7956	■ Production
fashion-clv-prediction	e9395aa4aaae...	R² = 0.9898	■ Production
fashion-segmentation	5b89ce0b1d58...	Sil = 0.2210	■ Production

CI/CD Pipeline — 7-step GitHub Actions

Step	Name	Tools	Trigger
1	Code Quality	flake8 · black · isort	Every push
2	Automated Tests	pytest (15+ cases)	Every push
3	Data Validation	Schema · null · leakage checks	Every push
4	Model Retraining	XGBoost · KMeans	Weekly cron
5	MLflow Registry	Metric-gated promotion	After training
6	Batch Scoring	All 8,000 customers	After training
7	Report Generation	Markdown + PDF auto-export	After scoring

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